ECE2800J

Programming and Elementary Data Structures

Introduction

Instructor

- Zhongqiang Ren
- Email: zhongqiang.ren@sjtu.edu.cn
- Phone: 3420-6045 (Ext. 5021)
- Office: Room 502, GC Building
- Office hour
  - During the break or after the lecture
  - Or by appointment via email

Time & Classrooms

Lectures:

- Tuesday10:00-11:40 am, DXY-315
- Wednesday 10:00-11:40 am, DXY-315
- Friday 8:55-11:40 am, DXY-315

Recitation:

Time and location will be notified on Feishu

Join Feishu Group

ECE2800J-SP2026

上海交通大学

Only members of this organization can join this group

This QR code is valid for 7 days (before 3/9)

Notes on Attending the Lectures

- Onsite teaching without online access
- There will be no recording
- Bring your laptop

References (Not Required)

- "C++ Primer, 4th Edition," by Stanley Lippman, Josee Lajoie, and Barbara Moo, Addison Wesley Publishing, 2005.
- "Problem Solving with C++, 8th Edition," by Walter Savitch, Addison Wesley Publishing, 2011.
- "Data Structures and Algorithm Analysis," by Clifford Shaffer.
  Online available:
  http://people.cs.vt.edu/~shaffer/Book/C++3e20120605.pdf

Teaching Assistants

- 1. Siqi Cao
- 2. Ziqian Wang
- 3. Yuxiang Yao

Communication

- Feishu
- Emails
- Canvas (Announcement)
- Onsite
  - During the break or after class

Grading

- Composition
  - Attendance Check: 5%
  - (About) 7 coding exercises 10%
  - 5 programming projects: 40%
  - Midterm exam (written): 20%
  - Final exam (written): 25%
- We will assign grades on a curve, in keeping with past grades given in this course.
- Questions about the grading?
  - Must be mentioned to TAs or instructor within one week after receiving the item.

Attendance Check

- Totally at random
  - Not sure which lecture
  - Not sure when (before, during, ending of the lec)
  - Don’t ask me: “Will there be attendance check this Wed?”

Attendance Check

- Totally at random
  - Not sure which lecture
  - Not sure when (before, during, ending of the lec)
  - Don’t ask me: “Will there be attendance check this Wed?”
- Scan the code
  - If failed, bring your student ID (or cell phone with your Siyuan code page) to me during the break or after the lecture.
  - **No way to re-check after the lecture**

Coding Exercises

- Get exercise on the knowledge you have learned
- Not too much coding
- Usually, one or two small problems each time

- **Some of them may appear as exam questions**

Projects

- Projects requirements:
  - Read and understand a problem specification
  - Design a solution (in your mind or on paper)
  - Implement this solution (simply and elegantly)
  - Convince yourself that your solution is correct

- **Some of them may appear as exam questions**

Grading of Exercises and Projects

- Grading coding exercises and projects will be done by a combination of testing (correctness) and reading (implementation requirement and simplicity/elegance).
- You will have chance to pre-test your program before the deadline.
  - We will use an online judge.
  - Pre-test cases are a subset of final test cases.
- For projects, we will give you a few simple test cases to get started. You should design your own set of tests (very important!).

Programming Environment

- We require you to develop your programs on **Linux operating systems** using compiler g++.
- C++17 standard is allowed.
  - Compile with the option -std=c++17
- We will grade your programs in the Linux environment.
  - They must compile and run correctly on this operating system.

Coding Exercise Deadline

- Each coding exercise will be given a due date. Your work must be turned in by 11:59 pm on the due date to be accepted for full credit

- No late submission allowed

Project Deadline

- Each project will be given a due date. Your work must be turned in by 11:59 pm on the due date to be accepted for full credit
- However, we still allow you to submit your homework within 3 days after the due date, but there is a late penalty

Hours Late	Scaling Factor
(0, 24]	80 %
(24, 48]	60 %
(48, 72]	40 %

- No work will be accepted if it is more than 3 days late!

Deadline Extension

- In **very occasional** cases, we accept deadline extension request.
  - Deadline extension requests will only be considered if you contact the course instructor in person. Do not contact TAs!
  - ONLY granted for documented medical or personal emergencies that could not have been anticipated.
  - NOT granted for reasons such as accidental erasure/loss of files and outside conflicting commitments.

Some Suggestions

- Practice! Build demos yourself
  - You have the freedom. Even try something wrong on purpose
- Learn from your mistakes!
  - Take notes on the mistakes you make. Review frequently
- Start your project/coding exercise early!
  - Don’t wait until the last minute. Numerous lessons before
  - **Hofstadter’s Law:** It always takes longer than you expect, even when you take into account Hofstadter’s Laws
- Make copies frequently in case your computer crashes.
  - Consequence: “computer crash” is NOT a reason for late submission!
  - (Version control tools such as git or svm, private)

Honor Code: Collaboration and Cheating

- You may discuss in oral with your classmates.
- **<u>But you must do all the assignments yourself</u>**.
- Some behaviors that are considered as cheating:
  - Reading another student’s answer/code, including keeping a copy of another student’s answer/code.
  - Copying another student’s answer/code, in whole or in part.
  - Having someone else write part of your assignment.
  - Using test cases of another student.
  - Testing your code with another one’s account.

“Another student” includes a student in the current semester or in the previous semester.

Honor Code: Collaboration and Cheating

- The previous lists of behaviors are **deliberate** cheating, but some **unintentional** actions could make you look like cheating. For example,
  - You use another’s computer to upload your code (in some cases like network/computer problems), but upload another’s copy.
- You should be extremely careful!
  - If due to network/computer problem, you need to use another’s computer, double check the uploaded file.

Honor Code: Collaboration and Cheating

- In summary, you should be responsible for all answers/codes you submit. If you submit a copy of another student’s work (or overwrite another student’s work), it is considered cheating, **no matter of the reason**!

Honor Code: Teaching and Learning Materials

- Teaching and learning materials, such as lecture slides, assignments, **your solutions**, quizzes, etc. are copyrighted and may not be passed on to others without the permission of the course instructor.
  - In particular, it is not permissible to post lecture slides, assignment questions, assignment solutions, etc., on public sites such as SlideShare
  - If you use Github to back up your code, make your repository **private**
  - You cannot use large language model (LLM)-based service, e.g., GPT, DeepSeek.

Consequence of Honor Code Violation

- Any suspect of honor code violation will be reported to **the Honor Council at GC**.
- For programming assignments, we will run an automated test to check for unusually similar programs. Those that are highly similar – in whole or in part – will be reported to **the Honor Council at GC**.
- **Penalty** of honor code violation

1. Reduction of the grade for this assignment to 0, **plus**
2. Reduction of the final grade for the course by one grade point, e.g., B+ → C+, for **both students** involved

Decision on a Violation of the Honor Code

November 2nd, 2024

[redacted]

		Term	Spring 2024
Instructor	Zhongqiang Ren	Course	ECE2800J Programming and Elementary Data Structures

The Honor Council has found that the student violated the Honor Code by improperly collaborating with another student on Projects 3 and 5.

The Faculty Committee on Discipline has therefore decided to impose the following sanctions:

1) Reduction of the points for Projects 3 and 5 to zero.
2) Reduction of the final grade for the course by two full grade points.

The course instructor and the Undergraduate Education Office will perform the appropriate modifications.

This decision may be appealed to the Dean's Cabinet of JI within 14 days of its receipt, following the stipulations of Section 6.12 of the Honor Code. Appeals should be addressed to the Associate Dean for Undergraduate Education.

Horst Hohberger

Chair of the Faculty Commission on Discipline

Canvas

- Log into Canvas: https://oc.sjtu.edu.cn/
- Check the class webpage on Canvas regularly for
  - Announcements
  - Slides
  - Grades
- Course slides will be uploaded onto Canvas before each lecture.

Getting Help

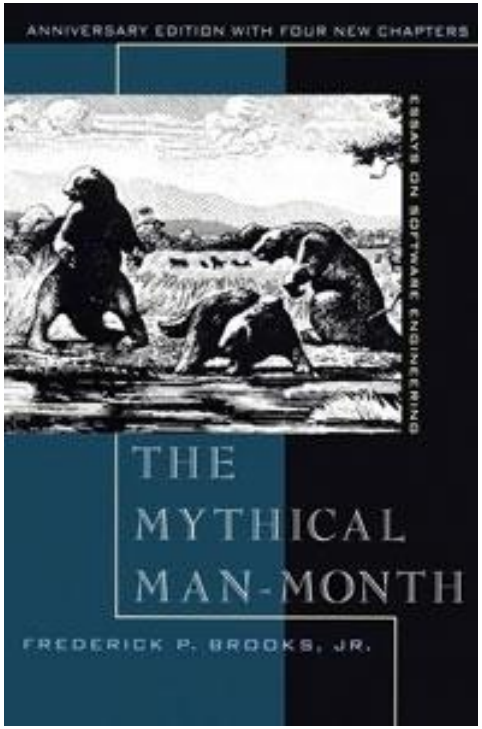
- If you have any questions, you can come to see TAs and instructor during the office hour
  - Better choice for questions that are not easy to solve!
- You can also post it on **Feishu**
  - You can help answer your fellow students' questions in the group chat
- For private question, you can also write emails to us

Aside: Fun Quizzes!

- What?
  - Multiple-choice questions on slides with
  - **Non-graded** and **Anonymous**
  - Feel free to answer even if you’re not sure!
- How?
  - Scan a QR code on your smartphone
  - Answer
  - Note: Some have a single answer; some can have more than one corrent answer
- Why?
  - Have fun!
  - Allow you to check your understanding
  - Allow the instructor to adapt his teaching
- Let’s try one!

Why is Programming Fun?

- First is the sheer joy of making things.
- Second is the pleasure of making things that are useful to other people.
- Third is the fascination of fashioning complex puzzle-like objects of interlocking moving parts and watching them work in subtle cycles.
- Fourth is the joy of always learning, which springs from the nonrepeating nature of the task.
- Finally, there is the delight of working in such a tractable medium.



The Mythical Man-Month: Essays on Software Engineering

What I Assume You Know

- Some basics of C++
  - Variables
  - Built-in data types, e.g., int, double, etc.
  - Operators, e.g., +, -, *, etc.
  - Flow of controls, e.g., if/else, while, for, switch/case, etc.
  - Functions; function declaration versus definition.
  - Arrays
  - Pointers
  - References
  - Struct

The Task of Programming

- Accept some specifications of the problem. (E.g., find the shortest way to go from my home to school.)
- Problem solving phase:
  - Design an algorithm (perhaps in pseudo-code/flow chart) that

  1) correctly satisfies the specification.
  2) is efficient in its usage of **space** and **time**.
- Implementation phase:
  - Implement the algorithm **correctly** and **efficiently**

  1) An implementation of an algorithm is correct if it behaves as the algorithm is intended for all inputs and in all situations.
     Correctness is never negotiable!
  2) **Efficient** can mean fast, simple, and/or elegant.

Problem Solving Phase

- Usually, hierarchical design: decompose into sub-tasks
- Example: find the shortest path from home to school

Pseudocode

```
void main() {
    graph_t map;
    node_t home, school;
    path_t path;
    (map, home, school) = read(filename);
    path = short_path(map, home, school);
    print(path);
}
```

Step 1: read the graph

Step 2: find the shortest path

Step 3: output the result

Key Points of ECE2800J

- The focus of ECE2800J is on the **implementation** part. Some **key points** we will learn include
  - Abstraction and its realization mechanism
  - Techniques to increase code reuse
  - Techniques to efficiently use memory
  - Elementary data structures
  - Some other essential parts of C++ programming

Abstraction

- One important concept about programming
  - Provides only those details that matter
  - Eliminates unnecessary details and reduces complexity
  - You already know one realization of abstraction: function (e.g. exp(x)), which is procedural abstraction

Abstraction

- We will talk about
  - Basics about abstraction
  - Procedure abstraction (i.e., function), in more detail
  - Data abstraction (i.e., class)
    - Basics about class: constructor, destructor, etc.
    - Abstract base class

Elegant code!

```
void main() {
    graph_t map;
    node_t home, school;
    path_t path;
    (map, home, school) = read(filename);
    path = short_path(map, home, school);
    print(path);
}
```

Data abstraction

Procedure abstraction

Techniques to Increase Code Reuse

- Function and class, which are basic ways to increase code reuse
- Class inheritance and virtual function
- Template and polymorphism
  - Template: write one thing, used for many different types

Techniques to Efficiently Use Memory

- Sometimes, the amount of memory needed to solve a problem can vary a lot
- Of course, you can write your program considering the worst-case memory usage
  - For example, a large enough array to hold data
  - However, this may lead to some waste in memory use

- We will learn a solution: **dynamic memory management**
  - Dynamic memory allocation and de-allocation

Elementary Data Structures

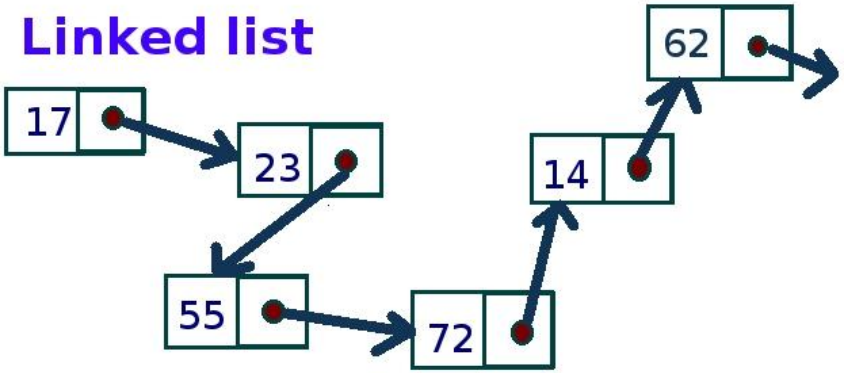
- Data structures are concerned with the **representation** and **manipulation** of data.
- All programs manipulate data.
- So, all programs represent data in some way.

Example: Store a list of numbers

Array

17	23	55	72	14	62

Linked list



Elementary Data Structures

- We will learn
  - Linked list
  - Linear list
  - Stack
  - Queue

- <u>**Note**</u>: This course only shows a few elementary data structures
  - More data structures will be taught in a following course, ECE2810J Data Structures and Algorithms

Other Essential Parts

- Writing programs that take arguments
- I/O streams, including file I/O
- Error handling
- Testing
- Linux

Good Programming Style

Meaningful Naming

Comments

```
// Evaluate the polynomial on x
int poly_eval(int x, int *coef, unsigned degree) {
      int result = 0;
      int x_power = 1;
      for(unsigned i = 0; i <= degree; i++) {
            result += coef[i] * x_power;
            x_power *= x;
      }
      return result;
}
```

Indentation

Consistency!

Relation with Other Courses

- ENGR1010J Introduction to Computers and Programming
  - Very basic programming skills.
  - ECE2800J will go in depth. To connect, we will review some basics.

- ECE2810J Data Structures and Algorithms
  - Focus on the efficiency of the algorithms.
  - ECE2800J focuses on correctness. It will show you some very basic data structures.

Recitation 1 !

Installing
Linux!

Questions?